

GovVerify AI

Unsupported Government Claim Detector

Problem ID: GOV-16

1. Project Title

Unsupported Government Claim Detector

GovVerify AI is an evidence-grounded verification system that checks AI-generated answers to government-related queries against official government documents. Instead of accepting a generated answer at face value, the system breaks it into individual factual claims, retrieves matching evidence from a curated repository of official documents, and classifies each claim as Supported, Uncertain, or Unsupported, along with the exact source document and page number backing the decision.

2. Introduction

Government departments and citizen-service platforms increasingly rely on AI assistants to answer questions about schemes, eligibility, procedures, and public services. These assistants can produce fluent, convincing responses, but generative models are prone to hallucination — stating facts that sound plausible but are not actually backed by any authoritative source.

Factual reliability matters far more in the government domain than in casual AI use. A citizen who acts on an incorrect eligibility rule, an incorrect document requirement, or a fabricated deadline can lose access to a benefit, miss an application window, or make a poor decision based on misinformation that appeared authoritative.

Unsupported AI claims in this domain create real risk: loss of citizen trust in government AI tools, spread of misinformation that is hard to trace back to its source, and difficulty for officials to audit what an AI system actually said and why. Official government documents — notifications, scheme guidelines, circulars, and manuals — remain the most reliable ground truth available, because they are published, versioned, and legally authoritative.

GovVerify AI exists to close this gap. Its purpose is to sit between an AI-generated government answer and the citizen or official reading it, verifying every factual claim against retrieved official evidence and making the basis of each claim transparent and traceable.

3. Problem Statement

AI systems can generate convincing government-related answers that may contain unsupported or hallucinated claims. A user reading such an answer generally has no way to tell which parts are grounded in an official source and which parts are not, because generated text does not carry its own evidence trail.

The proposed system should:

1. Accept a government-related query or a generated answer.
2. Identify individual factual claims within that answer.
3. Retrieve relevant evidence from official government documents.
4. Compare each claim with the retrieved evidence.
5. Classify each claim as Supported, Uncertain, or Unsupported.
6. Display the supporting evidence together with its source document and page number.

By grounding every claim in retrievable, citable evidence, GovVerify AI turns an opaque generated answer into an auditable one.

4. Proposed System

4.1 System Overview

GovVerify AI verifies AI-generated government information using a Retrieval-Augmented Generation (RAG) pipeline combined with a claim-level Natural Language Inference (NLI) verification stage. Official government documents are processed, chunked, and embedded into a searchable vector index. When a query is answered, individual claims are extracted from the generated response, matched against semantically relevant document chunks, and each claim-evidence pair is evaluated by an NLI model to decide whether the evidence entails, contradicts, or is insufficient to judge the claim.

The core building blocks of this system are:

- Government documents as the evidentiary source of truth
- Document processing and text chunking
- Semantic embeddings for meaning-based search
- Retrieval-Augmented Generation (RAG) for evidence-grounded answer generation
- Natural Language Inference for claim-evidence comparison
- Evidence matching with source and page-level citation
- Verification result generation and reporting

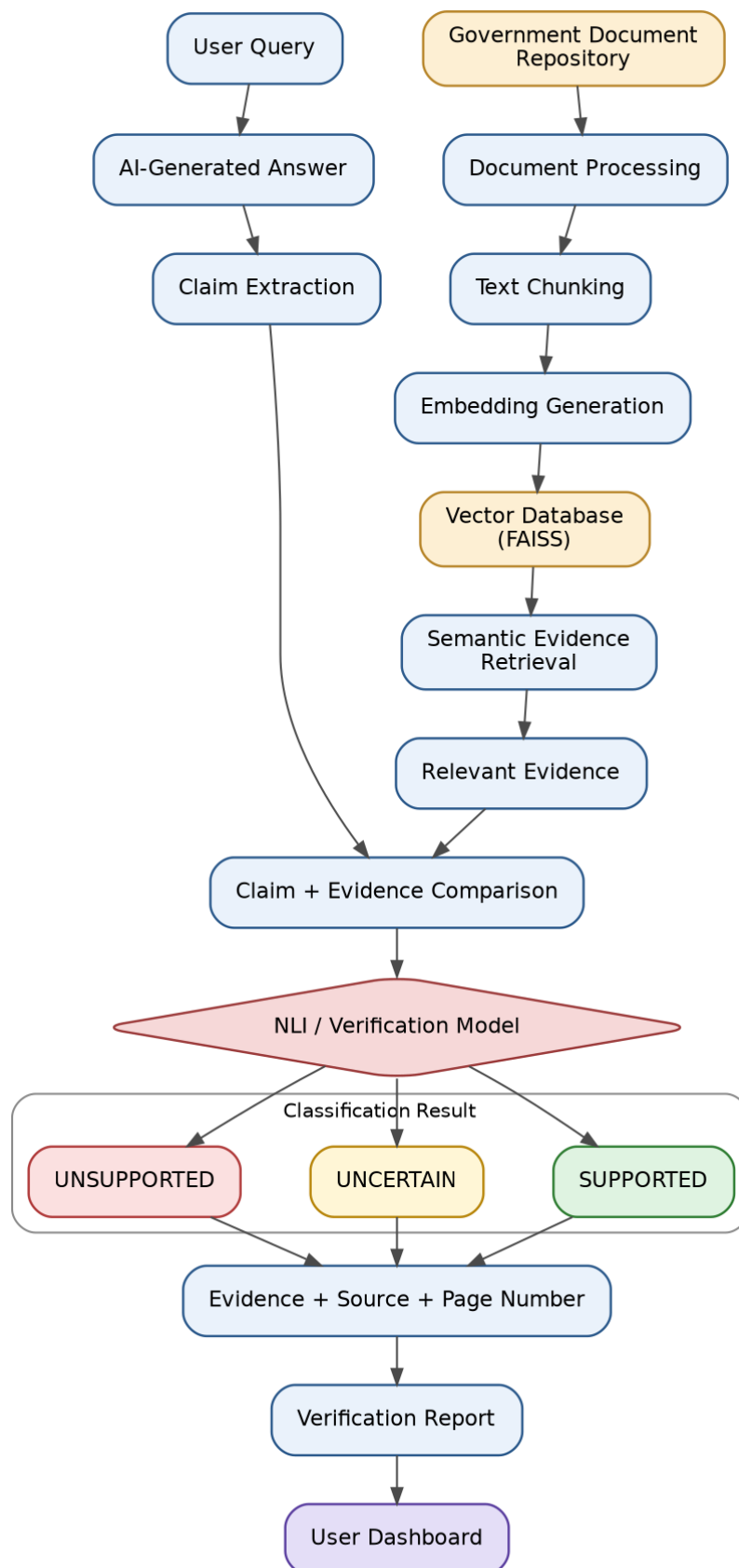
4.2 Key Features

- Government document ingestion
- PDF text extraction
- Semantic document search
- AI answer generation
- Claim extraction
- Claim-evidence matching
- Supported / Uncertain / Unsupported classification
- Evidence display
- Source document citation
- Page number citation
- Confidence / relevance scoring
- Verification report generation
- User-friendly dashboard

5. Objectives

- Detect unsupported claims in AI-generated government responses.
- Identify potentially hallucinated information.
- Ground AI responses using authoritative government sources.
- Retrieve relevant evidence using semantic search.
- Verify individual claims against retrieved evidence.
- Classify claims as Supported, Uncertain, or Unsupported.
- Provide page-level evidence and source references.
- Improve explainability and transparency of AI responses.
- Improve user trust in AI-generated government information.
- Maintain traceability between claims and original sources.
- Generate a structured and user-friendly verification report.

6. System Workflow



7. Technology Stack

Layer	Technology	Purpose
Frontend	React.js / TypeScript	Builds the interactive query and verification dashboard with type-safe components.
UI	Tailwind CSS	Provides consistent, utility-based styling for a clean, responsive interface.
Backend	Python / FastAPI	Serves REST endpoints for query handling, claim verification, and reporting.
PDF Processing	PyMuPDF	Extracts text and page metadata from official government PDF documents.
Embedding Model	BAAI/bge-m3	Converts text chunks and queries into semantic vector representations.
Vector Search	FAISS	Performs fast similarity search over document embeddings to retrieve evidence.
NLP Verification	DeBERTa-based NLI	Evaluates whether retrieved evidence entails, contradicts, or is insufficient for a claim.
Generative AI	LLM	Generates evidence-grounded answers and assists with claim extraction.
Database	PostgreSQL	Stores users, documents, verification requests, claims, and evidence records.
API Communication	REST API	Connects the frontend, backend, and verification services.
Version Control	Git / GitHub	Tracks source code changes and supports team collaboration.
Deployment	Vercel + Render / Railway	Hosts the frontend and backend services respectively.

8. Target Users

8.1 Citizens

People who want to verify whether an AI-generated answer to a government-related question is actually backed by an official source before acting on it.

8.2 Government Service Users

Users looking for information about:

- Government schemes
- Eligibility
- Benefits
- Required documents
- Application procedures
- Rules and guidelines

8.3 Government Officials

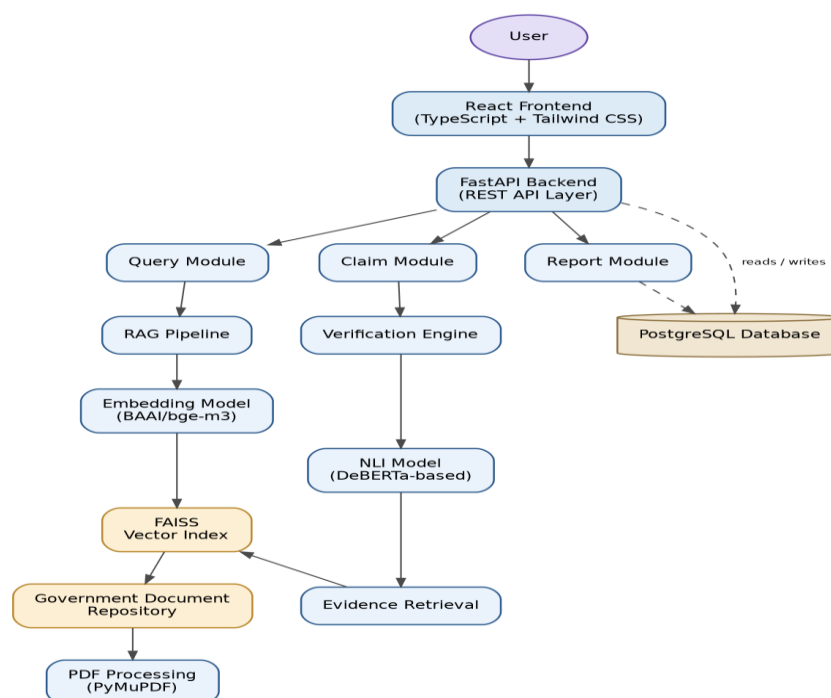
Officials who need to validate AI-generated public-service information before it is shared with citizens or published on official channels.

8.4 AI Researchers / Developers

Researchers and developers interested in:

- Retrieval-Augmented Generation (RAG)
- Hallucination detection
- Claim verification
- Evidence-grounded AI systems

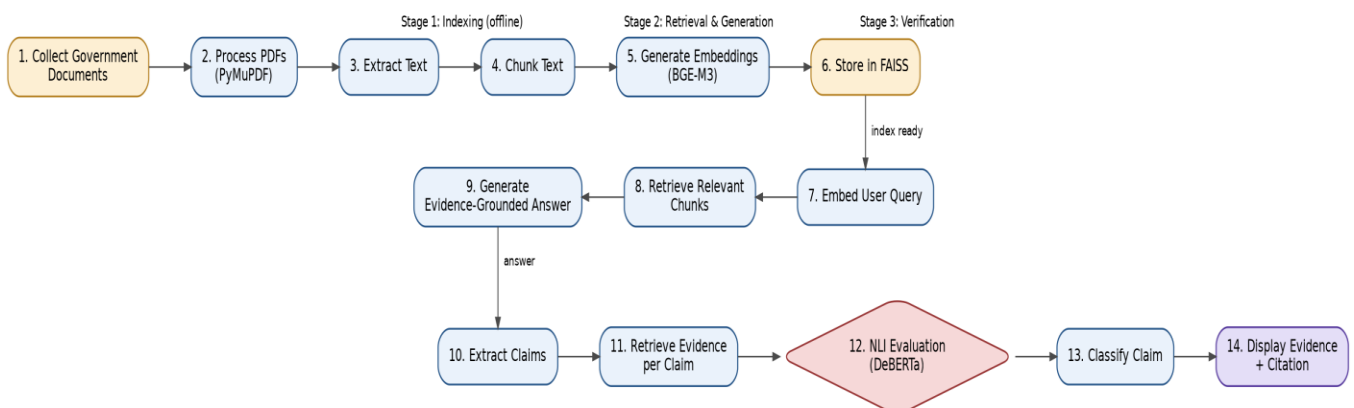
9. System Architecture



10. AI / RAG Pipeline

The end-to-end pipeline runs in three stages: offline indexing of government documents, retrieval-augmented answer generation, and claim-level verification.

1. Government documents are collected.
2. PDFs are processed.
3. Text is extracted.
4. Text is divided into meaningful chunks.
5. Embeddings are generated using BGE-M3.
6. Embeddings are stored in FAISS.
7. The user query is converted into an embedding.
8. Relevant document chunks are retrieved.
9. The AI generates an evidence-grounded answer.
10. Claims are extracted from the generated answer.
11. Evidence is retrieved for each individual claim.
12. The DeBERTa-based NLI model evaluates the claim-evidence relationship.
13. The claim is classified based on the NLI outcome.
14. Evidence and citations are displayed to the user.



11. Claim Classification Logic

Supported

The retrieved official evidence clearly supports the claim.

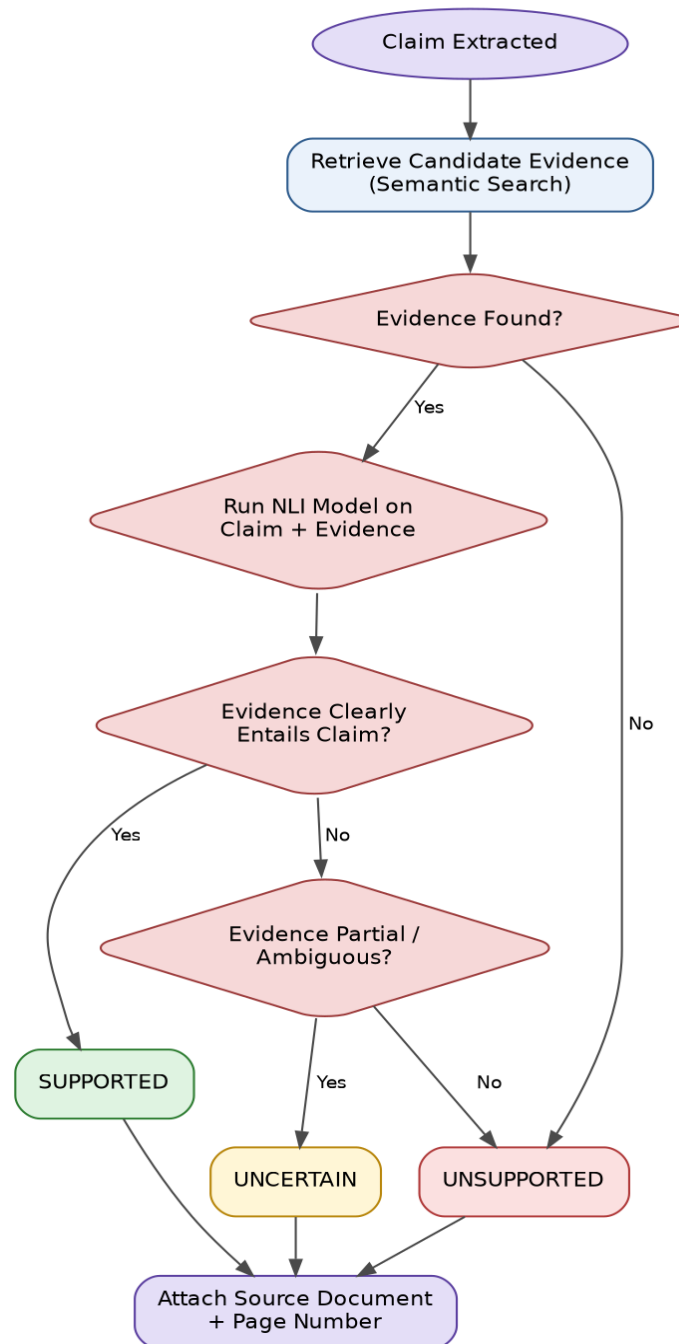
Uncertain

The evidence is incomplete, ambiguous, weak, or does not provide enough information to confidently verify the claim.

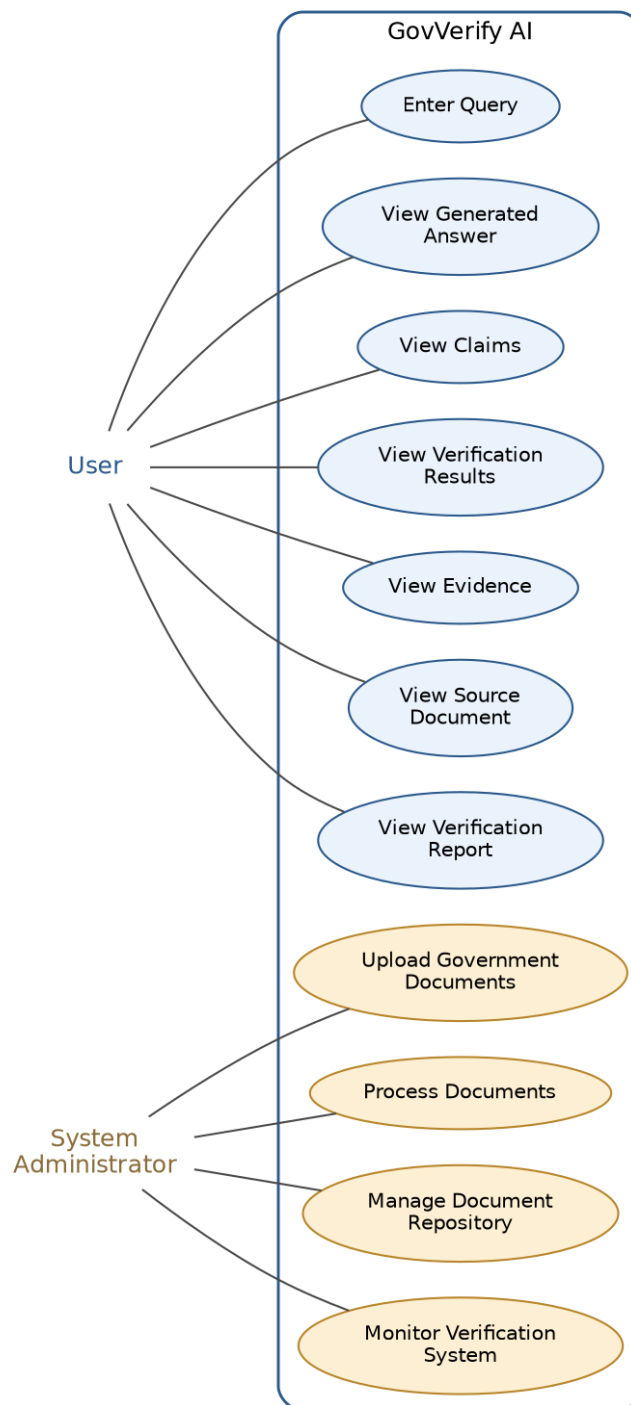
Unsupported

The system cannot find sufficient official evidence supporting the claim.

Important: "Unsupported" is never described as "False." The system is only checking whether sufficient retrieved official evidence supports the claim — it does not independently establish that the claim is factually wrong.



12. Use Case Diagram



The User can enter a query, view the generated answer, view extracted claims, view verification results, view supporting evidence, view the source document, and view the overall verification report. The System Administrator uploads government documents, triggers document processing, manages the document repository, and monitors the verification system.

13. Expected Output

The system produces a structured verification result summarizing every claim in the generated answer, its classification, and its evidence.

Sample Verification Report Output

Metric	Value
Total Claims	5
Supported	3
Uncertain	1
Unsupported	1

Sample Per-Claim Verification Detail

Claim	Status	Evidence
Claim 1	✓Supported	Official Government Document — Page 4
Claim 2	⚠Uncertain	Insufficient evidence found
Claim 3	XUnsupported	No supporting official evidence found

14. Advantages

- Detects unsupported AI claims before they reach the citizen
- Uses official government evidence as the source of truth
- Evidence-based, not opinion-based, verification
- Transparent, auditable results
- Page-level citations for every supported or uncertain claim
- Reduces the practical risk of AI hallucination in government information
- Easy for citizens to understand at a glance
- Scalable to multiple government domains and document sets
- Useful as a trust layer for existing government AI assistants

15. Future Enhancements

- Live official government website crawling
- Multilingual verification
- Tamil language support
- Voice-based queries
- Browser extension
- Government chatbot integration
- Advanced citation verification
- Cross-document contradiction detection
- Historical document comparison
- Real-time document updates
- Explainable AI verification
- More advanced hallucination detection models

16. Expected Impact

- Improves trust in AI-generated government information by making its basis visible
- Helps citizens verify information themselves instead of relying on blind trust
- Reduces the risk of misinformation caused by AI hallucinations in the government domain
- Improves transparency of government AI assistants that adopt this verification layer
- Encourages the broader adoption of evidence-grounded AI systems in public-sector tools

17. Conclusion

GovVerify AI provides an evidence-grounded approach to verifying AI-generated government information. By extracting individual claims, retrieving matching evidence from official government documents, and classifying each claim as Supported, Uncertain, or Unsupported with a precise source and page citation, the system turns an opaque generated answer into a transparent, auditable one. This gives citizens, officials, and AI developers a practical, reusable trust layer for government AI assistants, directly addressing the risk of unsupported or hallucinated claims in a domain where factual accuracy matters most.